

WASHINGTON HARBOR, DC, USA

WMO: 997314

Lat: 38.873N

Lon: 77.021W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.9	(c) -6.3	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 10.6	(k) 0.6	(l) 9.3	(m) -0.4	(n) 4.2	(o) 310	(p) 0.405

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.4	(c) 34.1	(d) N/A	(e) 32.6	(f) N/A	(g) 31.4	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 3.2	(p) 320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.7	(b) 7.5	(c) 6.6	(e) -10.6	(f) 36.1	(g) 2.9	(h) 1.9	(i) -12.7	(j) 37.4	(k) -14.4	(l) 38.5	(m) -16.0	(n) 39.6	(o) -18.2	(p) 41.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	1.9	4.2	8.1	15.1	20.5	24.7	27.1	25.8	22.5	15.9	9.6	5.2		
	DBStd	9.53	4.91	5.04	4.99	4.20	3.89	2.94	2.54	2.17	3.13	3.95	3.99	4.06		
	HDD10.0	744	253	171	97	7	0	0	0	0	0	4	54	158		
	HDD18.3	2131	508	395	319	115	24	1	0	0	3	93	263	408		
	CDD10.0	2608	3	9	37	159	326	441	531	489	376	188	41	8		
	CDD18.3	953	0	0	1	17	92	192	272	231	129	19	0	0		
	CDH23.3	8128	0	0	7	111	579	1616	2899	2003	829	84	1	0		
	CDH26.7	2755	0	0	0	21	139	533	1189	636	219	17	0	0		
(14) Wind	WSAvg	2.7	3.1	3.1	3.1	3.0	2.6	2.6	2.5	2.3	2.2	2.5	2.6	2.9		
(15) Precipitation	PrecAvg	1017	71	67	85	74	97	98	101	90	93	82	75	82		
	PrecMax	1684	181	144	215	175	271	356	247	233	314	239	192	166		
	PrecMin	722	10	9	1	1	19	24	17	9	3	0	7	11		
	PrecStd	194	33	35	39	34	51	72	54	52	70	53	45	41		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.0	20.3	24.2	29.6	31.9	34.9	36.7	34.3	32.6	28.7	21.3	18.3		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	14.2	16.5	20.7	26.4	29.9	32.7	34.9	32.5	31.0	25.5	19.0	15.7		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	11.5	13.5	18.2	24.0	28.1	31.1	33.3	31.3	29.3	23.5	17.3	13.8		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	8.8	11.1	15.7	22.1	26.6	29.7	31.9	30.0	27.7	21.7	15.7	11.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		6.9	7.7	8.1	9.2	8.3	7.6	7.4	7.2	7.3	7.4	7.4	6.3		
	5% DB	MCDBR	10.6	11.6	12.0	11.8	9.5	9.0	8.7	8.4	8.6	9.8	10.1	9.8		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.309	0.323	0.358	0.404	0.445	0.502	0.523	0.518	0.430	0.363	0.325	0.311		
	taud		2.488	2.430	2.396	2.286	2.203	2.083	2.042	2.040	2.261	2.453	2.517	2.519		
	Ebn at Noon		869	902	900	874	841	790	769	761	813	845	843	841		
	Edh at Noon		79	95	109	129	143	162	167	164	124	92	76	72		
(44) All-Sky Solar Radiation	RadAvg		2.08	2.93	3.89	4.92	5.55	6.13	6.01	5.36	4.37	3.28	2.37	1.83		
	RadStd		0.17	0.33	0.28	0.37	0.62	0.47	0.31	0.28	0.51	0.36	0.20	0.18		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (75 neighbors)		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+124	+72

Nomenclature: See separate page